

STUDIFY

Use-Case Specification: Mark Study Status

Version: 1.0

Revision History

Date	Version	Description	Author
23/07/2026	1.0	Initial version of Mark Study Status Use-Case Specification	Thiên Phước

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1. Use-Case Name

1.1 Brief Description

This use case describes the process by which a User self-assesses and records their knowledge of a flashcard after reviewing it during a study session. After flipping a card to view its explanation (UC8 – Flip Card to View Explanation), the User rates how well they knew the answer by selecting a **study status** (e.g., "Know", "Don't Know", "Review Later"). The system records this rating and uses it to update the flashcard's study progress, which may inform a spaced repetition algorithm to schedule future reviews. This use case is included by UC7 (Study Flashcards).

2. Flow of Events

2.1 Basic Flow

This use case starts immediately after the User has flipped a flashcard to view its Explanation during a study session (upon completion of UC8 – Flip Card to View Explanation).

1. The system displays the **Mark Study Status** controls below or alongside the flipped flashcard. The controls consist of three buttons:
 -  **"Know"** (green): The User knew the answer confidently.
 -  **"Don't Know"** (red): The User did not know the answer.
 -  **"Review Later"** (yellow/amber): The User partially knew the answer or wants to review again soon.
 2. The User selects **"Know"** to indicate they knew the term and its explanation confidently.
 3. The system records the **"Know"** status for this flashcard in the current study session.
 4. The system updates the flashcard's study metadata:
 - Increments the "correct" count for this card.
 - Schedules the next review date according to the spaced repetition algorithm (the next review is set further in the future for well-known cards).
 - Marks the card as "mastered" if it has been marked "Know" a sufficient number of consecutive times (configurable threshold).
 5. The system advances to the next flashcard in the study session deck (returning control to UC7 – Study Flashcards, Step 8).
 6. The use case ends for the current card.
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2.2 Alternative Flows

2.2.1 User Marks "Don't Know"

This alternative flow describes the behavior when the User did not know the flashcard's term or explanation.

1. After viewing the Explanation (Step 1 of the Basic Flow), the User clicks the  **"Don't Know"** button.
2. The system records the **"Don't Know"** status for this flashcard.
3. The system updates the flashcard's study metadata:
 - Increments the "incorrect" count for this card.
 - Resets or shortens the spaced repetition interval, scheduling the card for an earlier re-review (e.g., within the same session or the next day).
 - Adds this card to the "Difficult" or "Re-review" pile of the current session.
4. The system may re-insert this card into the session queue (e.g., it will appear again near the end of the current session).
5. The system advances to the next new card in the deck.
6. The flow returns to Step 5 of the Basic Flow (advance to next card).

2.2.2 User Marks "Review Later"

This alternative flow describes the behavior when the User partially knew the answer or wants a second look.

1. After viewing the Explanation (Step 1 of the Basic Flow), the User clicks the  **"Review Later"** button.
2. The system records the **"Review Later"** status for this flashcard.
3. The system updates the flashcard's study metadata:
 - Does not increment the correct count.
 - Schedules the card for an intermediate re-review date (shorter interval than "Know" but longer than "Don't Know").
 - Adds the card to a secondary review queue within the same session (it will reappear once near the end).
4. The system advances to the next card in the primary deck.
5. The flow returns to Step 5 of the Basic Flow (advance to next card).

2.2.3 User Uses Keyboard Shortcut to Mark Status

This alternative flow describes keyboard-based status marking for accessibility and efficient study.

1. After the card has been flipped (UC8 completed), the User presses one of the designated keyboard shortcuts:
 - **Key "1" or Arrow Right** → Marks **"Know"**.
 - **Key "2" or Arrow Down** → Marks **"Review Later"**.
 - **Key "3" or Arrow Left** → Marks **"Don't Know"**.
2. The system recognizes the keypress and processes it as the equivalent button click.
3. The system records the status and the flow continues as per the corresponding Basic or Alternative flow above.

2.2.4 User Changes a Previously Marked Status

This alternative flow occurs within a session when the User wants to change the status they marked for the most recently reviewed card.

1. After marking a status and being shown the next card, the User clicks an **"Undo"** button (if provided) within a 3-second grace period.
2. The system cancels the advancement and returns the User to the previously marked card.
3. The system un-records the previously selected status.
4. The system re-displays the Mark Study Status controls for the returned card.
5. The User selects a new status.
6. The flow resumes at Step 3 of the Basic Flow (or corresponding alternative flow) with the new status.

3. Special Requirements

3.1 Usability Requirements

- The three status buttons must be large, clearly labeled, and color-coded to minimize the cognitive load during a study session.

- The buttons must only appear after the card has been flipped (i.e., after UC8 is triggered), to prevent premature marking without reviewing the explanation.
- The current card's marked status must be visually reflected immediately upon selection (e.g., a brief highlight/animation on the chosen button).

3.2 Performance Requirements

- The status must be recorded and the next card must be loaded within 300 milliseconds of the User pressing a status button.
- Study status updates must be saved to the backend asynchronously (non-blocking) to maintain a smooth session experience.

3.3 Spaced Repetition Algorithm Requirements

- The system must implement a spaced repetition scheduling algorithm (e.g., SM-2 or a simplified variant) based on the User's marked status history.
- The next review date for each card must be calculated and stored after each status marking.
- Cards marked "Don't Know" must be prioritized in future sessions.

3.4 Data Persistence Requirements

- All study status records must be durable: once a status is recorded and the User advances, the data must not be lost even in the event of a network interruption (use local storage as a fallback).

4. Preconditions

4.1 Active Study Session

- The User must be in an active flashcard study session (UC7 – Study Flashcards).

4.2 Card Has Been Flipped

- The flashcard must have been flipped (UC8 – Flip Card to View Explanation must have been completed) before the Mark Study Status controls are made available. The controls must not be accessible before the card is flipped.

4.3 User Authentication

- The User must be authenticated (logged in) to the STUDIFY system.

5. Postconditions

5.1 Status Recorded

After the use case is completed successfully:

- The selected study status ("Know", "Don't Know", or "Review Later") is recorded for the current flashcard within the current study session.

- The flashcard's overall study metadata (correct count, incorrect count, review interval, next review date) is updated accordingly.
- The session progresses to the next flashcard.

5.2 Session Summary Impact

At the end of the study session:

- The system's session summary (displayed after the last card in UC7) reflects the breakdown of cards by marked status (e.g., 15 "Know", 3 "Don't Know", 2 "Review Later").

5.3 Long-term Study Progress

After persistent saving:

- The User's study progress for each flashcard is updated in the system, influencing the spaced repetition schedule for that card.
- Cards consistently marked "Know" multiple times may be labeled as **"Mastered"** and deprioritized in future sessions.

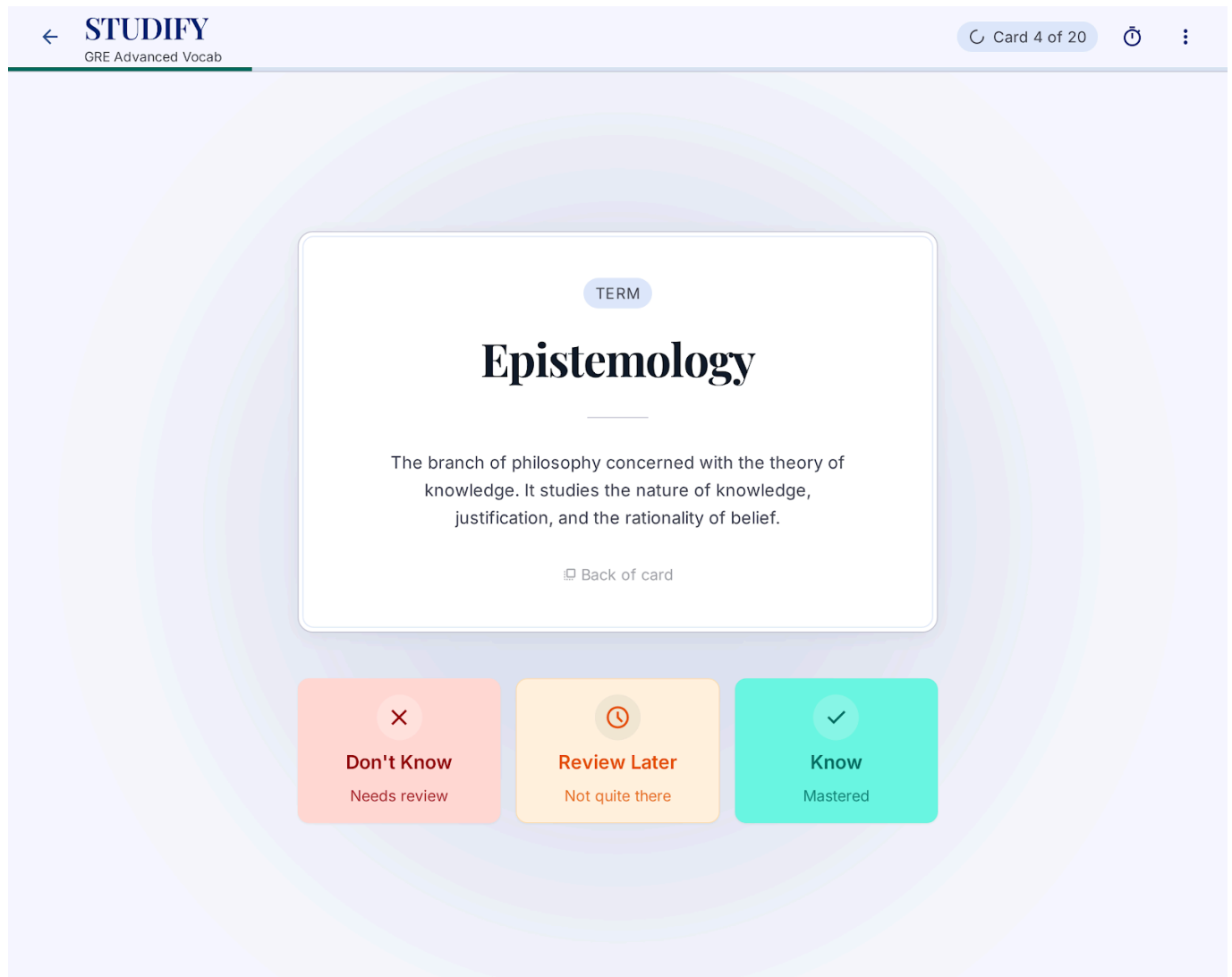
6. Extension Points

This use case has no extension points of its own. It is a mandatory component included by UC7 (Study Flashcards), and is triggered after UC8 (Flip Card to View Explanation) completes for each card during a study session.

7. UI Prototype

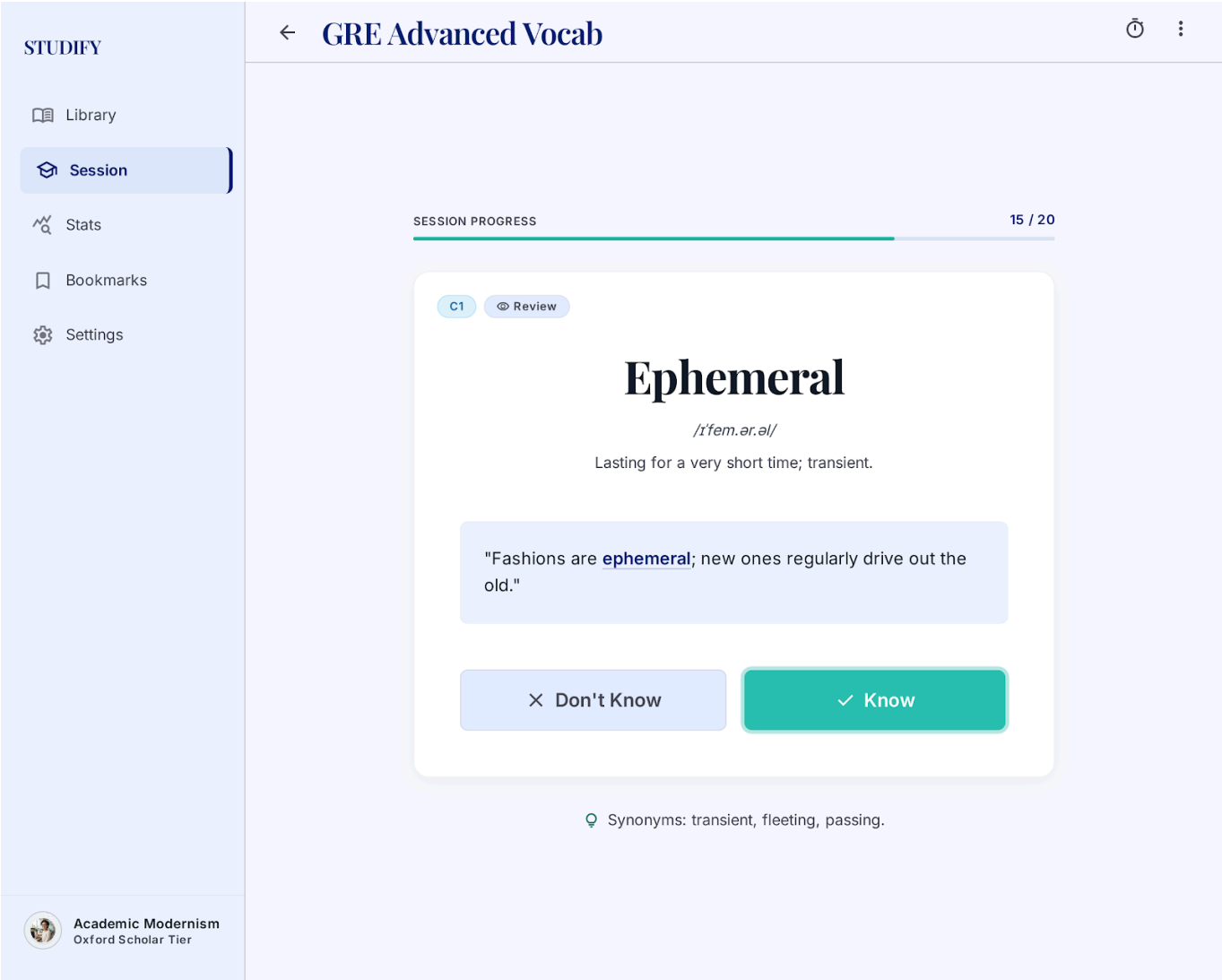
7.1 Basic Flow — Status Buttons Displayed After Flip

Shows the three status buttons (✅ Know / 🔄 Review Later / ❌ Don't Know) visible after the card is flipped.



7.2 Basic Flow — "Know" Button Selected

Shows the Know button highlighted green after the User selects it.



7.3 Alternative Flow 2.2.1 — "Don't Know" Selected

Shows the Don't Know button highlighted red; card will be re-queued.

← **STUDIFY** **GRE Advanced Vocab**  

Card 12 of 45

26%

VOCABULARY

Obfuscate

"The writer deliberately used complex terminology to _____ the true meaning of the text."

 /əb'fʌs.keɪt/


Don't Know
< 1m


Hard
10m


Good
1d


Easy
4d

 Card added back to today's queue for re-review. 

7.4 Alternative Flow 2.2.2 — "Review Later" Selected

Shows the Review Later button highlighted amber.

The screenshot displays the STUDIFY application interface. At the top, there is a header bar with a close button (X) and the word 'STUDIFY' on the left, and a progress indicator '14 / 50' with a green bar and a menu icon (three dots) on the right. The main content area features a large title 'GRE Advanced Vocab' in a dark blue serif font, with the subtitle 'Philosophy & Methodology' in a smaller, lighter blue sans-serif font below it. A central card with a light yellow background and rounded corners contains the word 'Epistemology' in a large, bold, black serif font. Above the word is a small blue pill-shaped button with the word 'Noun' in white. Below the word is a definition: 'The theory of knowledge, especially with regard to its methods, validity, and scope, and the distinction between justified belief and opinion.' At the bottom of the card is a quote: '"His **epistemology** was grounded in empirical observation rather than rationalist deduction."' Below the card, there are three buttons: 'Review Later' (orange with a clock icon), 'Got It' (light blue with a checkmark icon), and 'Mastered' (light blue with a gear icon). At the bottom of the screen, a dark blue notification bar with rounded corners contains an information icon (i) and the text 'Card marked for review in next session.'

7.5 Alternative Flow 2.2.3 — Keyboard Shortcut Labels on Buttons

Shows keyboard shortcuts (1 / 2 / 3 or arrow keys) displayed on each status button.

The screenshot displays the 'GRE Advanced Vocab' application interface. On the left is a sidebar with navigation options: 'Library', 'Session' (highlighted), 'Stats', 'Bookmarks', and 'Settings'. The top header shows a back arrow, the title 'GRE Advanced Vocab', and icons for a clock and a menu. The main content area features a 'SESSION PROGRESS' bar at the top right indicating '12 / 50'. Below this is a large white card for the word 'Perspicacious', labeled 'C2' in the top right corner. The card contains the word in a large serif font, its definition in italics: *"having a ready insight into and understanding of things."*, and an example sentence: 'Example: The perspicacious student quickly grasped the complex nuances of the UI design system.' At the bottom of the card are three interactive buttons: '1 Know' with a green checkmark icon, '2 Review Later' with a clock icon, and '3 Don't Know' with a red X icon.

7.6 Alternative Flow 2.2.4 — Undo Toast (Grace Period)

Shows the brief undo notification allowing the User to revert the last marked status.

STUDIFY

GRE Advanced Vocab

Progress

5 / 20

Ephemeral

"The beautiful sunset was an ephemeral moment of joy."

lasting for a very short time.

Reveal Answer

Status saved as Know.

UNDO

11 / 11